HONG KONG BAPTIST UNIVERSITY
SCHOOL OF CONTINUING EDUCATION

Higher Diploma in Data Science
Teaching and Learning Plan

Semester:	2nd, 2025/26
Course Title and Code:	HDDS1502 Statistics for Data Science
Course Instructor:	Dr. Keith Lai
Course Duration:	13 weeks x 3 hours (lecture)
Consultation Hours:	Thursday morning (by appointment)
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Contact No.:	3411-5437
Level:	4

Course Aims:

This course aims to provide students with a conceptual understanding of statistics for data science. Topics include measures of central tendency and dispersion, tabular and graphical presentations of statistics, probability and probability distributions, sampling and hypothesis testing, correlation, regression and overfitting.

Course Intended Learning Outcomes (CILOs)

CILOs	**By the end of the course, students should be able to:**
CILO 1	Apply the basic principles of descriptive statistics in relation to various forms of data and data collection;
CILO 2	Calculate the occurrence of events with basic probabilities and probability distributions;
CILO 3	Analyse data using inferential statistics techniques; and
CILO 4	Design spreadsheet models for the analysis of different types of statistics.

Teaching and Learning Activities (TLAs)

CILOs	TLAs will include the following:
CILOs 1-4	• Interactive lectures supplemented with slides, reading notes, video clips and other supporting materials.
CILO 1	• Case studies of real-life situations to illustrate the applications of descriptive statistics. • Class discussion to analyse problems encountered in different situations and derive possible solutions.
CILO 2	• Use practical activities and experiments to examine the probabilities for the occurrence of everyday events. • Class discussion to formalise and translate a given situation into a probability model to predict possible outcomes.
CILO 3	• Demonstration using different inferential statistics techniques to illustrate the process of estimation and modelling from data. • Class discussion to clarify the applications of appropriate inferential statistics techniques using the examples provided.
CILO 4	Computer laboratory activities: • Demonstration of various skills and statistical techniques. • Completion of exercises by students to practise skills and techniques in statistics.

Assessment Methods (AMs)

Type of Assessment Methods	Weighting	CILOs to be addressed	Description of Assessment Tasks
Individual Assignment	25%	2, 3, 4	Students will complete two individual assignments on problem solving using inferential statistics techniques, such as correlation and regression analysis, on different sets of data.
Group Assignment	35%	1, 4	Students, in groups of 3 – 4, will be asked to collect real-life data and conduct statistical analysis using spreadsheets. Students will need to perform different statistical calculations and present their results in an appropriate format.

Final Examination	40%	1, 2, 3, 4	Students will sit for a 3-hour written examination, consisting of multiple-choice, short and long questions, on the following topics: central tendency and dispersion, probabilities, sampling and hypothesis testing, correlation and regression.

Statement about Generative AI Tools use in Assessed Assignment Tasks

Given the proliferation of generative AI tools and their utility in teaching and learning, in alignment with the University's relevant principles, the default approach is that students are empowered and guided to use generative AI tools in assessed assignment tasks. Such use must align with the University's academic integrity guidelines for students while ensuring they are achieving the PILOs and CILOs.

The "Principles for the Use of Generative AI Tools in Teaching and Learning, and Assessment" ("Principles") guides us to use generative AI tools for better learning outcomes. You may access the "Principles" on BUniPort (U-Wide Policies & Info -> Policies, Guidelines & Standards -> Quality Assurance, Teaching & Learning).

In this course, generative AI tools can be used in the assignment tasks in the specific way mentioned:

- explain or clarify concepts
- planning and brainstorming on projects
- grammar check for the write-up

Assessment Brief

Task name:	Individual Assignments
Task description:	Students will complete two individual assignments on problem solving using inferential statistics techniques, such as correlation and regression analysis, on different sets of data.
Task weight:	25%
Word limit:	Not applicable
Submission date:	**12 Feb (part 1), 19 Mar (part 2)**

Task name:	Group Assignment
Task description:	Students, in groups of 3 – 4, will be asked to collect real-life data and conduct statistical analysis using spreadsheets. Students will need to perform different statistical calculations and present their results in an appropriate format.
Task weight:	35%
Word limit:	**Not applicable**
Submission date:	**9 Apr**

Task name:	Final Examination
Task description:	Students will sit for a 3-hour written examination, consisting of multiple-choice, short and long questions, on the following topics: central tendency and dispersion, probabilities, sampling and hypothesis testing, correlation and regression.
Task weight:	40%
Time required:	3 hours
Examination date:	**TBD**

Course Schedule

Week	Content	CILOs	Assessments
1 (15/1)	Introduction to Statistics • Introduction to descriptive statistics, probabilities and inferential statistics • Statistics in data science	CILOs 1, 2, 3	
2 (22/1)	Descriptive Statistics • Types of data and methods of data collection • Measures of central tendency • Measures of dispersion	CILO 1	
3 (29/1)	Descriptive Statistics with Spreadsheets • Handling data with spreadsheets • Data presentation with spreadsheets • Statistical functions in spreadsheets • Lab Practice I on descriptive statistics	CILOs 1, 4	
4 (5/2)	Correlation and Regression • Correlation analysis and correlation coefficient • Linear regression	CILO 3	
5 (12/2)	Correlation and Regression • Least squares method • Multivariable regression and overfitting	CILO 3	**Individual Assignment (part 1 due)**
6 (26/2)	Regression with Spreadsheets • Linear regression with spreadsheets • Multivariable regression with spreadsheets • Lab Practice II on correlation and regression	CILOs 3, 4	
7 (5/3)	Introduction to Inferential Statistics • Differences between descriptive and inferential statistics • Sampling distributions • Central Limit Theorem	CILO 3	
8 (12/3)	Hypothesis Testing (I) • Basic introduction • Hypothesis testing with one population	CILO 3	
9 (19/3)	Hypothesis Testing (II) • The null and alternative hypotheses	CILO 3	**Individual Assignment**

	• Types of errors		**(part 2 due)**
10 (26/3)	Introduction to Probabilities • Types of probabilities • Conditional probability • Counting principles • Decision tree	CILO 2	
11 (2/4)	Probability Distributions • Basic concepts of probability distributions • The binomial probability distribution	CILO 2	
12 (9/4)	Probability Distributions • The binomial probability distribution • The normal probability distribution	CILO 2	**Group Assignment (due)**
13 (16/4)	Course Review	CILOs 1-4	

Remark: The teaching schedule may vary depending on the classes' progress.

References

Textbooks

Berenson, M.L., Levine, D., Szabat, K. & Stephan, D. (2024). *Basic business statistics: concepts and applications* (15th ed.). Pearson.

Recommended Readings

Donnelly, R.A., Abdel-Raouf, F. (2016). *Statistics*. Alpha, Penguin Random House.

Forsyth, D. (2018). *Probability and statistics for computer science*. Springer.

Oakshott, L. (2020). *Essential quantitative methods for business, management and finance* (7th ed.). Palgrave Macmillan.

Quirk, T.J. (2020). *Excel 2019 for business statistics: a guide to solving practical problems* (2nd ed.). Springer.

Skiena, S.S. (2017). *The data science design manual*. Springer.